

Compiler Design
Midsemester Assignment
Spring 2020-2021

1. Write down the output for the following input expression: $p = x / y * (y - z) + 15$ in equivalent assembly code, where all variables are float. Show the steps you used to produce the assembly code. 7
2. Construct a syntax directed translation scheme that translates arithmetic expressions from infix notation into postfix notation. Show the application of your scheme to the string $5 / 4 * 2 + 3$. 7
3. Add 'Semantic Rules' and 'Translation Schemes' to the following productions of a grammar. 6

Productions	Semantic Rules	Translation Schemes
$S \rightarrow U$		
$U \rightarrow T a U$		
$U \rightarrow T a T$		
$T \rightarrow a T b T$		
$T \rightarrow b T a T$		
$T \rightarrow d$		

Instructions

1. Submission Date: **March 12,2021**. (you must submit the assignment before 11.59 PM).
2. Document Type: Hand Written -> Scanned-> PDF format
3. Write down each question first then the answer to the question sequentially as numbered.
4. **DO NOT COPY, if found then you (also who helped you to copy the assignment) all of you will get zero.**